

MambaRefine-CD: Region-Boundary Temporal Refinement with MambaVision for Remote Sensing Change Detection

Abstract—Binary remote-sensing change detection requires spatially coherent change masks and precise boundary delineation. Existing CNN and Transformer-based Siamese architectures often fuse temporal evidence into a single shared representation, which can limit recovery of fine object boundaries. This paper presents MambaRefine-CD, a MambaVision-based binary change detection framework that explicitly decouples temporal evidence into complementary region-aware and boundary-aware responses. A shared MambaVision encoder extracts aligned multi-scale features from bi-temporal image pairs. The proposed Differential Region–Boundary Interaction (D-RBI) module constructs a temporal descriptor from paired features, absolute difference, and signed difference, and routes it through separate gated region and boundary streams before multi-scale aggregation. An adaptive receptive-field FPN decoder aggregates the region features, and a bounded residual refinement head adjusts coarse predictions near boundaries using the finest boundary stream. Experiments on DSIFN-CD and WHU-CD show that the proposed model is competitive with recent Mamba-based change detection methods while providing explicit region–boundary temporal refinement. A controlled ablation over seven variants isolates the contribution of each component.

I. INTRODUCTION

Remote-sensing change detection (CD) estimates a binary change mask from two co-registered images acquired over the same area at different times. The task underpins urban expansion monitoring, disaster assessment, land-cover analysis, and infrastructure management. In practice, binary CD is difficult because the model must compare two observations while being invariant to nuisance variation from illumination shifts, seasonal changes, shadows, sensor differences, and small registration errors. These factors are particularly visible near object boundaries, where a model can correctly identify the approximate changed region and still produce a poor mask if the predicted boundary is shifted, fragmented, or over-smoothed.

Deep CD methods have evolved along three main directions. Convolutional Siamese networks compare bi-temporal images through early fusion or feature differencing, as in FC-EF, FC-Siam-conc, and FC-Siam-diff [1]. Later CNN encoder–decoder models improve localization through skip connections, dense feature reuse, and multi-scale attention [2]–[4]. These models are effective because convolutions provide useful locality and translation bias, but their context modeling is bounded by the receptive field, which can limit reasoning about spatially distributed changes. Transformer-based methods address long-range interaction more directly: BIT models bi-temporal to-

kens with a Transformer encoder [5], and ChangeFormer uses a hierarchical Siamese Transformer for dense CD [6]. Wijenayake et al. [7] further demonstrate that precise spatio-temporal feature fusion yields gains over standard differencing strategies. Despite the progress, attention-based models carry high memory and compute cost at dense resolution, and stronger global representations do not automatically resolve boundary localization errors.

Recent visual state-space models (SSMs) provide a different route. Mamba introduces selective state-space sequence modeling with linear complexity [8], and VMamba adapts it to vision through two-dimensional selective scanning [9]. MambaVision develops a hybrid Mamba–Transformer backbone for visual recognition and dense prediction tasks [10]. Mamba-based CD models, including ChangeMamba [11], HAM-CD [12], and Mamba-CD [13], show that SSM backbones are competitive for binary and semantic change detection. Wijenayake et al. in Mamba-FCS [14] further demonstrate that combining frequency-domain fusion with change-guided attention and a SeK-inspired loss improves semantic CD. However, applying a Mamba-based encoder alone is not a complete solution. A CD model must still construct temporal evidence from both images and preserve fine boundaries in the output mask. A controlled benchmark of SSM backbones for remote-sensing segmentation by Wasalathilaka et al. [15] found that boundary delineation remains the dominant failure mode under domain shift, suggesting that encoder scaling alone is insufficient for boundary-sensitive tasks.

We propose **MambaRefine-CD**, a MambaVision-based architecture for binary remote-sensing CD. The model builds temporal features using raw paired features, absolute difference, and signed difference, giving the decoder access to both magnitude-based and direction-aware evidence. The core novelty is **D-RBI**, which applies two separate learned gating mechanisms to temporal features before decoding: a region gate that produces stable, coherent changed-area responses, and a Sobel-conditioned boundary gate that selects contour-sensitive evidence. These two pathways are combined through an adaptive receptive-field decoder and refined by a bounded residual head that applies a controlled correction to coarse logits near boundaries.

The contributions of this paper are:

- We present **MambaRefine-CD**, a MambaVision-based binary CD architecture integrating shared bi-temporal

encoding, explicit four-stream temporal feature construction, and boundary-aware refinement.

- We propose **D-RBI**, a lightweight module that decouples temporal evidence into complementary region-aware and boundary-aware gated pathways before multi-scale decoding.
- We integrate adaptive receptive-field decoding, CRAM-lite spatial modulation, and bounded residual boundary refinement into a unified pipeline, demonstrating that each component contributes measurably.
- We report verified results on DSIFN-CD and WHU-CD, averaged over independent runs, and present a controlled ablation study over seven variants (A0–A6) that isolates each component’s effect on F1, IoU, and OA.

II. RELATED WORK

A. Convolutional and Transformer-Based Change Detection

Early deep learning CD models used convolutional Siamese encoders with feature differencing or concatenation as the temporal comparison mechanism [1]. Dense decoders with skip connections and multi-scale aggregation improved spatial localization [2]–[4]. These CNN-based approaches are efficient and translation-equivariant, but their effective receptive field limits the ability to relate spatially distant changed regions.

Transformer-based methods address global context more directly. BIT encodes bi-temporal features as semantic tokens and applies cross-attention for temporal reasoning [5]. ChangeFormer uses a hierarchical Siamese Transformer with an MLP decoder for dense prediction [6]. Ratnayake et al. show that incorporating attention and improved loss design into segmentation architectures benefits change detection tasks [16]. While transformers improve global modeling, their quadratic attention complexity makes high-resolution binary CD computationally expensive, and improved global representations do not directly translate to better boundary delineation.

B. State Space Models for Remote Sensing

Mamba introduces selective SSM sequence modeling with linear complexity [8]. VMamba extends this to vision via two-dimensional selective scanning [9]. MambaVision develops a hybrid Mamba–Transformer backbone with competitive accuracy and efficiency on dense prediction benchmarks [10].

In CD, ChangeMamba integrates VMamba into a bi-temporal architecture and demonstrates competitive performance [11]. HAM-CD introduces a hybrid attention mechanism to compensate for Mamba’s limitations in fine-grained local modeling [12]. Wijenayake et al. combine spectral-domain fusion with change-guided attention and a SeK-inspired loss in Mamba-FCS for semantic CD [14], while Wijenayake et al. further explore precise spatio-temporal fusion strategies for binary CD [7].

Peng et al. proposed Mamba-CD [13], a VMamba-based binary change detection network with change region-aware

attention and a gated recursive context refinement mechanism. Their work is closely related to MambaRefine-CD because it also studies Mamba-style long-range modeling and boundary-sensitive binary change detection. However, Mamba-CD enhances shallow features using a change-region attention map and recursively fuses multiscale context, whereas MambaRefine-CD explicitly constructs temporal evidence using paired features, absolute difference, and signed difference, and separates region and boundary streams through D-RBI before bounded boundary residual refinement.

These works demonstrate the viability of SSM backbones for remote-sensing CD, but they primarily address region-level detection and do not explicitly separate boundary-sensitive cues from region-level evidence.

C. Boundary-Aware Modeling and Temporal Feature Construction

Boundary precision is a well-recognized challenge in segmentation. Boundary loss directly penalizes errors at object contours [17]. Gated-SCNN processes a dedicated shape stream in parallel with the main segmentation path [18]. BAS-Net uses a two-stage coarse-then-fine architecture for salient object boundary detection [19]. In remote sensing, boundary-aware multi-task learning has been studied for aerial dense prediction [20], and BSCNet uses boundary-aware semantic context to sharpen CD edge predictions [21].

A key limitation of most CD methods, including recent Mamba adaptations, is that they compare bi-temporal features through absolute differencing $|F_2 - F_1|$ alone. Absolute differencing is symmetric and discards the temporal direction of the change (e.g., whether a structure was built or demolished). Wasalathilaka et al. confirm that boundary delineation remains the dominant failure mode of SSM backbones under domain shift, even at larger scales [15]. MambaRefine-CD addresses both issues: it preserves signed temporal direction through $F_2 - F_1$ and explicitly gates the temporal feature into region and boundary pathways before decoding.

III. METHODOLOGY

A. Overview

MambaRefine-CD predicts a binary change map from a bi-temporal image pair (I_1, I_2) through five components: a shared MambaVision encoder, a four-stream temporal feature constructor, D-RBI, an adaptive receptive-field FPN decoder with CRAM-lite modulation, and a bounded boundary residual refinement head. As shown in Fig. 1, these components form an end-to-end pipeline for binary CD.

B. Shared MambaVision Encoding

A shared encoder $E(\cdot)$ processes both input images in the same feature space:

$$\{F_1^s\}_{s=1}^S = E(I_1), \quad \{F_2^s\}_{s=1}^S = E(I_2). \quad (1)$$

Weight sharing ensures that the temporal difference computed downstream reflects genuine change rather than encoder domain mismatch. We use MambaVision-S as the default backbone (channels [96, 192, 384, 768], $S = 4$ stages), initialized

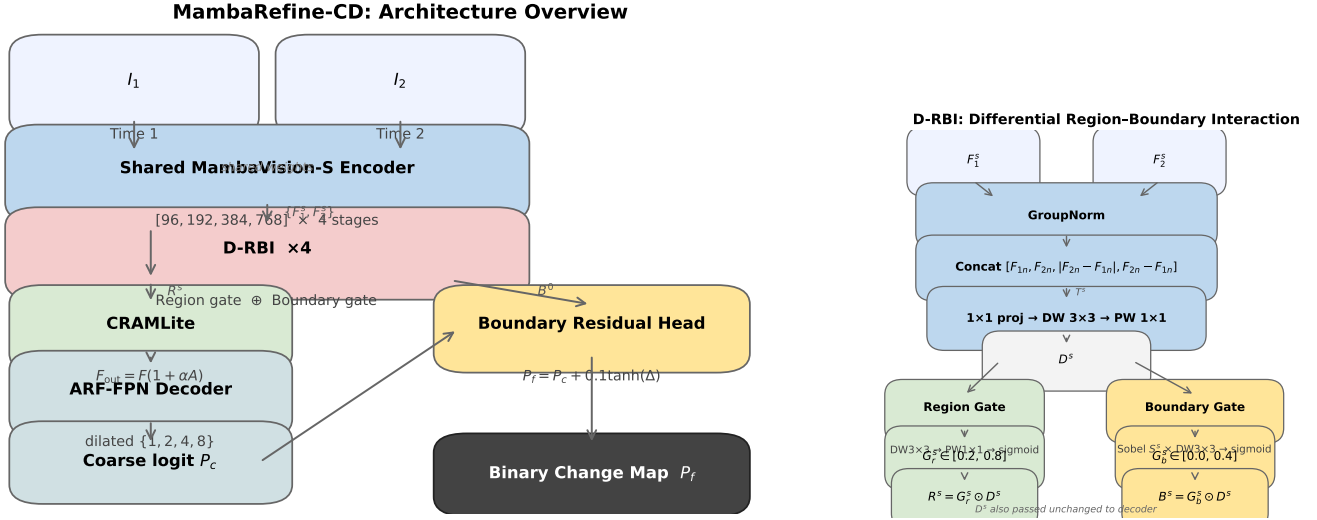


Fig. 1. (Left) Overall architecture of MambaRefine-CD: a shared MambaVision encoder extracts multi-scale bi-temporal features; per-scale D-RBI blocks produce region and boundary streams; ARF-FPN with CRAM-lite decodes region features; a bounded residual head refines the coarse prediction. (Right) D-RBI structure: temporal descriptor D^s (from paired features, absolute difference, and signed difference) is routed through a bounded region gate and a Sobel-conditioned boundary gate.

from pretrained weights and kept trainable. The backbone fuses Mamba layers for long-range modeling and Transformer layers for local attention [10].

C. Four-Stream Temporal Feature Construction

At each encoder stage s , we apply independent GroupNorm to F_1^s and F_2^s , then concatenate four streams:

$$T^s = [F_{1,n}^s, F_{2,n}^s, |F_{2,n}^s - F_{1,n}^s|, F_{2,n}^s - F_{1,n}^s], \quad (2)$$

where $F_{i,n}^s$ denotes the normalized feature. Absolute difference provides a symmetric change-magnitude cue; signed difference preserves the temporal direction of change, which conveys whether a structure appeared or disappeared. This four-stream concatenation T^s is projected to a fixed channel width through a lightweight 1×1 convolution followed by a depthwise 3×3 and pointwise 1×1 block, yielding the temporal descriptor D^s .

D. Differential Region-Boundary Interaction (D-RBI)

D-RBI applies two separate gating mechanisms to D^s to expose region-level and boundary-level evidence before decoding. The D-RBI module is detailed in Fig. 1 (right).

Region gate. A spatial attention path processes D^s through a depthwise 3×3 convolution and a pointwise 1×1 convolution to produce unnormalized gate logits, which are clamped to $[-8, 8]$ and passed through sigmoid, then scaled to bound the response:

$$G_r^s = 0.6 \cdot \sigma(\text{clamp}(g_r(D^s), -8, 8)) + 0.2, \quad G_r^s \in [0.2, 0.8]. \quad (3)$$

The lower bound prevents complete feature suppression; the upper bound prevents unrestricted amplification.

Boundary gate. A Sobel-conditioned gate modulates its response using the magnitude of image-level edges computed

on D^s . Sobel magnitude is clamped to $[0, 10]$ and normalized, providing a spatial prior on likely boundary locations:

$$G_b^s = 0.4 \cdot \sigma(S^s \cdot g_b(D^s)), \quad G_b^s \in [0.0, 0.4], \quad (4)$$

where S^s denotes the normalized Sobel magnitude map. The tighter bound keeps the boundary gate from dominating when edge evidence is unreliable.

Outputs. The two gated responses and the original descriptor are returned:

$$R^s = G_r^s \odot D^s, \quad B^s = G_b^s \odot D^s, \quad D^s \text{ (unchanged)}. \quad (5)$$

These three streams give the decoder access to region evidence, boundary evidence, and unmodulated temporal features.

E. CRAM-Lite Modulation and Adaptive Receptive-Field Decoding

At selected encoder stages ($s \in \{0, 1, 2\}$), a lightweight CRAM-lite block reweights the region features spatially. Following the motivation of change-region-aware attention in Mamba-CD [13], each CRAM-lite block estimates a per-location attention map $A \in [0, 1]$ via depthwise and pointwise convolutions and applies a residual modulation:

$$F_{\text{out}} = F_{\text{region}} \cdot (1 + \alpha \cdot A), \quad \alpha \in \mathbb{R}^+ \text{ (learnable, init 0.5)}. \quad (6)$$

The modulated multi-scale features are decoded through a top-down FPN with adaptive receptive-field blocks. Each block runs four parallel dilated depthwise convolutions with rates $\{1, 2, 4, 8\}$, weights each branch via per-image softmax attention (estimated through global average pooling and a shared linear layer), and sums the weighted outputs. This gives the decoder access to both compact object-level changes and larger spatial structures without altering the input resolution.

F. Coarse Head and Bounded Boundary Residual Refinement

The decoder first produces a coarse logit map P_c through a 3×3 convolution followed by a 1×1 output projection, bilinearly upsampled to the input resolution. A boundary refinement head then computes a bounded correction:

$$P_f = P_c + 0.1 \cdot \tanh(\Delta), \quad (7)$$

where Δ is estimated from the concatenation of the coarsest boundary feature B^0 , the coarse logit P_c , and Sobel edges of $\sigma(P_c)$. The $\tanh$ bounds the correction magnitude; the scalar 0.1 limits refinement to uncertain edge regions without freely overwriting the coarse global prediction. The final linear layer of the boundary head is zero-initialized to ensure the model begins training from the coarse prediction alone.

G. Training Objective

The full training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + 0.4 \mathcal{L}_{\text{coarse}} + 0.1 \mathcal{L}_{\text{boundary}}, \quad (8)$$

where $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{Dice}}$ supervise the final output P_f ; $\mathcal{L}_{\text{coarse}}$ applies the same binary supervision to P_c ; and $\mathcal{L}_{\text{boundary}}$ is a Sobel-edge ℓ_1 loss between predicted and ground-truth boundaries. Ablation variants A0–A5 are trained with only $\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$ on the final output (coarse and boundary loss weights set to zero).

IV. EXPERIMENTS

A. Datasets and Split Protocol

We evaluate MambaRefine-CD on two widely used binary CD benchmarks.

DSIFN-CD [22] contains 394 Gaofen-2 image pairs (1 m GSD) covering six Chinese cities. We use the official image-level split (395 train, 48 val, 48 test), extract patches from training images only, and evaluate test images in patch mode with 25% overlap. We refer to this as the *clean split*.

WHU-CD [23] is an aerial building CD dataset over Christchurch, New Zealand. We use the official three-way split and evaluate in the same patch mode.

B. Implementation Details

Models are trained with AdamW ($\text{lr} = 5 \times 10^{-5}$, weight decay = 0.01), cosine scheduling with 2500 warmup steps, batch size 8, and 256×256 crops. Gradient norm is clipped at 0.5. Exponential moving average (EMA, decay = 0.999) weights are used for inference. Augmentation: random horizontal and vertical flips. Training uses 16-bit AMP on a single NVIDIA Quadro GV100 (32 GB). Ablation variants A0–A5 train for 30k iterations; the full model (A6) trains for 50k. The validation threshold is swept over $\{0.30, \dots, 0.60\}$ in 0.05 steps and fixed for test evaluation; test metrics are never used for selection.

TABLE I
COMPARISON ON WHU-CD. LITERATURE VALUES FROM [13]. PARAMS IN M; FLOPS NOT REPORTED FOR PRIOR METHODS UNDER COMPARABLE SETTINGS.

Method	Params	Pre	Rec	F1	IoU	OA
ChangeFormer [6]	–	91.83	88.02	89.88	81.63	99.12
BiFA	–	95.15	93.60	94.37	89.34	99.56
RSM-CD	–	93.37	90.42	91.87	84.96	–
CDMamba	–	95.58	92.01	93.76	88.26	99.51
SChanger	–	94.62	91.83	93.20	87.27	–
Mamba-CD [13]	–	96.52	93.91	95.20	90.83	99.62
MambaRefine-CD (ours)	65.40	95.58	94.74	95.15	90.76	99.54

C. Evaluation Metrics

We report Precision (Pre), Recall (Rec), F1, IoU, and Overall Accuracy (OA) for the changed class via a global confusion matrix. F1 and IoU are primary metrics; OA is dominated by the unchanged class and is reported for completeness. We additionally evaluate Boundary F1 (BF1), Boundary IoU (BIOU), and Trimap F1 at 3-pixel tolerance to provide direct evidence of boundary localisation quality.

D. Main Results

On the DSIFN-CD clean split, MambaRefine-CD achieves F1=95.67 and IoU=91.71 (averaged over two canonical runs) using 65.40M parameters. The best single run reaches F1=96.40 and IoU=93.04, indicating run-to-run variance is present. On WHU-CD, the model achieves F1=95.15 and IoU=90.76 at threshold 0.55.

E. Comparison with Recent Methods

We compare MambaRefine-CD with six representative binary CD methods spanning CNN, Transformer, and Mamba-based architectures. Literature values are taken from Peng et al. [13]. Colored values denote **best**, **second**, and **third** results per metric column.

WHU-CD (Table I): MambaRefine-CD is competitive with the recent Mamba-CD baseline [13], closely matching it in F1 and IoU while obtaining higher recall. No state-of-the-art claim is made.

DSIFN-CD (Table II): Literature methods use the Mamba-CD-style patch split (14 400/1 360/192 non-overlapping 256×256 pairs). Because our ablation runs use the clean image-level split, a direct same-protocol comparison is not yet available; the MambaRefine-CD row is marked [VERIFY]. We note that the clean-split result (F1=95.67, IoU=91.71) is close in magnitude to Mamba-CD (F1=95.61, IoU=91.69) on the Mamba-CD-style split; this is contextual rather than a same-protocol ranking, and no state-of-the-art claim is made on DSIFN-CD.

F. Ablation Study

Table III presents a controlled ablation over seven variants on DSIFN-CD. The largest single gain comes from replacing the SimpleCNN-FPN baseline (A0) with MambaVision-S (A1), confirming the dominant role of the backbone. The A2 row (unsigned D-RBI only) is a single verified run and

TABLE II
COMPARISON ON DSIFN-CD. LITERATURE VALUES FROM [13] UNDER THE MAMBA-CD-STYLE PATCH SPLIT. PARAMS IN M.

Method	Protocol	Params	Pre	Rec	F1	IoU
ChangeFormer [6]	Lit. patch	–	88.48	84.94	86.67	76.48
BiFA	Lit. patch	–	73.99	68.87	71.34	55.45
FTAN	Lit. patch	–	90.54	88.61	89.56	81.10
ADSFNet	Lit. patch	–	94.79	95.24	95.01	90.50
Mamba-CD [13]	Lit. patch	–	95.60	95.61	95.61	91.69
MambaRefine-CD (ours)	Lit. patch	65.40	[VERIFY]	[VERIFY]	[VERIFY]	[VERIFY]

Lit. patch = Mamba-CD-style 14400/1360/192 non-overlapping patch split.
[VERIFY] = same-protocol evaluation pending.

TABLE III
ABLATION ON DSIFN-CD. A0–A5 USE 30K ITES; A6 USES 50K.
VALUES AVERAGED OVER AVAILABLE RUNS. † SINGLE VERIFIED RUN.
PARAMS IN MILLIONS.

ID	Variant	Key Change	F1	IoU	OA	Params
A0	CNN baseline	SimpleCNN + FPN	76.63	62.12	83.80	7.84
A1	+MambaVision-S	Replace encoder	93.87	88.45	95.72	53.54
A2†	+D-RBI (abs)	Abs diff only	93.33	87.50	95.35	54.98
A3	+Signed diff	Add $F_2 - F_1$ stream	94.28	89.19	95.99	55.34
A4	+ARF-FPN	Add CRAM-lite	94.36	89.32	96.05	65.12
A5	+Bnd. head	Head, no bnd. loss	93.59	87.94	95.53	65.19
A6	Full model	+ CRAMLite + bnd. loss	95.67	91.71	96.98	65.40

should be interpreted cautiously; additional runs are left for future work. Adding signed temporal difference $F_2 - F_1$ (A3) increases F1 and IoU beyond A2, supporting the value of directional temporal evidence. ARF-FPN with CRAM-lite modulation (A4) provides a further small positive gain. Adding the boundary residual head alone without boundary-aware loss terms (A5) causes a regression (-0.77 F1), showing that the boundary head is not beneficial in isolation. The full model (A6) with CRAMLite, coarse auxiliary supervision, and boundary loss recovers and exceeds all prior variants, confirming that boundary refinement is most effective when trained jointly with the full loss.

G. Boundary-Specific Evaluation and Backbone Scaling

We also evaluate boundary-specific metrics (BF1, BIoU, Trimap F1 at 3-pixel tolerance) across ablation variants on DSIFN-CD. The full model (A6) achieves BF1=71.94 and Trimap F1-3px=72.93, compared to BF1=61.36 for A1 and BF1=65.94 for A4, confirming that joint boundary supervision measurably improves boundary localisation. For reference, the WHU-CD full model achieves BF1=89.49, reflecting the simpler and more uniform building boundary structures.

Regarding backbone scaling on DSIFN-CD: moving from MambaVision-T (46.73 M, F1=94.58, Pre=94.33, Rec=94.84) to MambaVision-S (65.40 M, F1=95.67) yields a clear gain. Moving to MambaVision-B (113.37 M, F1=95.77) adds only a marginal further gain at nearly double the parameter count, suggesting diminishing returns beyond MambaVision-S.

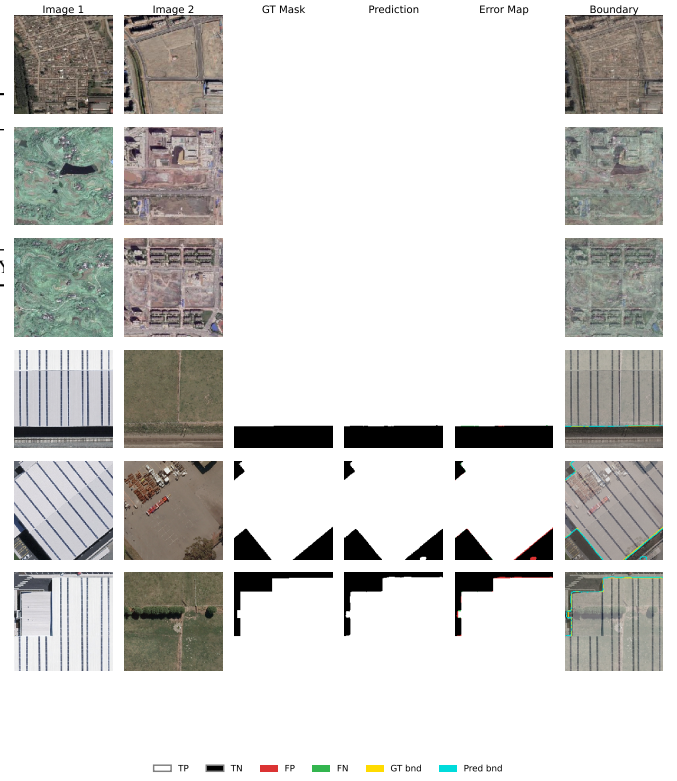


Fig. 2. Qualitative examples on DSIFN-CD (top rows) and WHU-CD (bottom rows). Each row: I_1 , I_2 , GT, prediction, error map (white = TP, black = TN, red = FP, green = FN), boundary overlay (yellow = GT bnd, cyan = pred bnd).

H. Qualitative Analysis

Qualitative examples in Fig. 2 illustrate the full model’s prediction quality on both datasets. The full model (A6) produces sharper boundary delineation with fewer false positives near edges, consistent with the boundary-specific metrics.

I. Limitations

The Sobel-based boundary gate provides a useful spatial prior but is fixed, not learned end-to-end. DSIFN-CD evaluation protocols differ across studies, making strict cross-paper comparisons unreliable without a shared split. A proper same-protocol evaluation under the Mamba-CD-style patch split for DSIFN-CD is pending.

V. CONCLUSION

This paper presented MambaRefine-CD, a MambaVision-based architecture for binary remote-sensing change detection. The method decouples temporal evidence into region-aware and boundary-aware pathways through D-RBI, combines signed temporal difference with adaptive receptive-field decoding and CRAM-lite spatial modulation, and applies a bounded residual correction near boundaries. A controlled ablation over seven variants demonstrates that the MambaVision encoder provides the dominant accuracy gain, while signed temporal difference, CRAM-lite modulation, and the combined boundary head plus Sobel-edge loss each contribute additional

improvement. On DSIFN-CD (averaged over two runs) the method achieves $F1=95.67$ and $IoU=91.71$. On WHU-CD, it achieves $F1=95.15$ and $IoU=90.76$. The results show that MambaRefine-CD is competitive with recent Mamba-based change detection methods, including Mamba-CD [13], while providing a different region-boundary temporal refinement design. Future work will extend evaluation to additional binary CD benchmarks, investigate learned boundary gate designs beyond the Sobel prior, and explore applying the region-boundary decomposition to semantic CD settings.

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